II Jacobson Radical Theory

4 The Jacobson Radical

Exercise 1.1: secretly Lemma 4.1 part a)

Let R be a ring. Define the Jacobson radical $J(R)$ as the intersection of all maximal left ideals. Prove that $J(R)$ is, in fact, a two-sided ideal. (As a lemma, prove that an element $x \in R$ belongs to the Jacobson radical if and only if $1 - axb$ is invertible for all $a, b \in R$.)

Exercise 2.1: secretly Lemma 4.1 part b)

Let R be a ring and let $r \in R$. Prove that r belongs to $J(R)$ iff $rM = \{0\}$ for all simple left ${}_RM$.

Exercise 3.1: secretly Theorem 4.14 part a)

A ring is said to be **semiprimitive** if $J(R) = \{0\}$. Prove that if R is a semisimple ring, then R is artinian and semiprimitive.

Proof.

Suppose R is semisimple. Then, there exists some left ideal I such that:

$$R \;=\; J(R) \oplus I$$

If $J(R) \neq (0)$, then $I \neq R$, and is thus contained in some maximal ideal I'. But then $I' \nsubseteq J(R)$, a contradiction. Thus, R must be semiprimitive. Since a ring is semisimple if and only if it is noetherian and artinian, we are done. ■

Exercise 4.1

Find $J(\mathbb{Z}_n)$ for all $n \in \mathbb{N}$.

Exercise 5.1

Let R, S be two rings. What is $J(R \times S)$?

Proof.

dupe of problem 8 below? ■

Exercise 6.1

Let R be a ring and let $n \geq 1$. What is $J(M_n(R))$?

Exercise 7.1

Let R be a ring with ideal I. What is $J(R/I)$?

Exercise 8.1: Exercise 4.12B

Let A, B be two rings. Show that $J(A \times B) = J(A) \times J(B)$.

Exercise 9.1: Exercise 4.21

Let $\mathrm{GL}_n(R)$ denote the group of units in the matrix ring $M_n(R)$. Suppose that I is an ideal of R contained in $J(R)$. Show that the map $\mathrm{GL}_n(R) \to \mathrm{GL}_n(R/I)$ is onto.

5 Jacobson Radical under Change of Rings

Exercise 10.1: Corollary 5.2

Let R be a commutative ring. Prove that the polynomial ring $R[x]$ is semiprimitive (meaning its Jacobson radical is zero) if and only if R is a reduced ring (meaning it has no nilpotents).

Exercise 11.1: Theorem 5.3

Prove that if k is a field and R a finitely-generated k-algebra, then R is semiprimitive.

Exercise 12.1: Proposition 5.6

Let R be a ring, and let A be a subring of R. Suppose that A is a direct summand of R (as a left R-module). Prove that $J(A) = J(R) \cap A$.

Exercise 13.1

Let R be a commutative ring. Prove that the intersection of all prime ideals of R is equal to the set of nilpotents in R.

Exercise 14.1

Let R be a commutative ring and let $p(x) = \sum_{i=0} a_i x^i$ be in $R[x]$. Prove that $p(x)$ is a unit in $R[x]$ if and only a_0 is a unit in R, and $a_1, \ldots, a_n$ are nilpotents in R.

Hint: show that $a_1, \ldots, a_n$ belong to every prime ideal of R. Thus use the previous exercise to deduce that $a_1, \ldots, a_n$ are nilpotent.

Exercise 15.1: Exercise 5.6

Let R be a ring, let $R[[t]]$ be the ring of formal power series in an indeterminate t with coefficients from R. Note that $tr = rt$ for all $r \in R$. Let (t) denote the ideal of $R[[t]]$ generated by t. This coincides with the set of power series with no constant term:

$$(t) := \{tf(t) : f(t) \in R[[t]]\} = \{f(t) \in R[[t]] : f(0) = 0\}$$

Prove that the Jacobson radical of $R[[t]]$ is:

$$J(R[[t]]) = J(R) + (t)$$

Exercise 16.1: secretly Theorem 4.14 part b)

Prove that if R is artinian and semiprimitive, then R is semisimple.

Proof.

Suppose R is artinian and semiprimitive. Note R must have the following two properties:

a) Every nonzero left ideal U contains a minimal left ideal B. This follows from the fact that R is artinian.

b) Every minimal left ideal B is a direct summand of ${}_RR$ with a maximal left ideal.

To see property b), note that since $J(R) = 0$, then there necessarily exists a maximal left ideal m such that $B \not\subseteq m$. Since $B \cap m$ is an ideal of B and B is minimal, then $B \cap m = (0)$. Since m is maximal, note that $B + m = {}_RR$, and therefore ${}_RR = B \oplus m$.

Now, take any minimal left ideal B_1 and, by b), write:

$$ {}_RR = B_1 \oplus U_1 $$

Then $U_1 \neq 0$, so by a), there exists a minimal left ideal $B_2 \subseteq U_1$. By b), B_2 is a direct summand of ${}_RR$ and thus U_1, so we can write:

$$ U_1 = B_2 \oplus U_2 \implies {}_RR = B_1 \oplus B_2 \oplus U_2 $$

Continuing ad infinitum, if R is not semisimple, then we necessarily always have that $B_{k+1} \subsetneq U_k$. In other words, we get an infinite descending chain:

$$ U_1 \supsetneq U_2 \supsetneq U_3 \supsetneq \cdots $$

This contradicts the fact that R is artinian, so we must have that R is semiprimitive as needed. ■

V Two Theorems

1 The Artin-Wedderburn Theorem

Exercise 1.1

Let R be a ring. Show that R is semisimple if and only if R is both artinian and semiprimitive.

Proof.

See exercises 3 and 16 of the Section 4/5 worksheet. Alternatively, see Theorem 4.14 in the book. ■

Exercise 2.1: Chinese Remainder Theorem

Let R be a ring and let I, J be two ideals in R. There is a natural map $R \to (R/I) \times (R/J)$. Prove the Chinese Remainder Theorem: if $I + J = R$ (we say that I, J are **comaximal**), then this natural map is surjective.

Proof.

Note that the natural map Φ is given by $x \mapsto (x_I, x_J)$, where $x_I := x + I$ is the image of x under the projection map for R/I, and similarly for x_J.

Let $(a_I, b_J) \in (R/I) \times (R/J)$. Since $I + J = R$, then there exists $x \in I$ and $y \in J$ such that $x + y = 1 \in R$. Write:

$$a_I = a + I \qquad b_J = b + J$$

where $a, b \in R$ are the "lifts" of a_I and b_J. Now, consider the element $ay + bx$. Behold:

$$\begin{aligned} ay + bx &= a(1-x) + bx = a - ax + bx = a + (b-a)x \\ ay + bx &= ay + b(1-y) = b - by + ay = b + (a-b)y \end{aligned}$$

Since $x \in I$ and $y \in J$, it follows that $\Phi(ay + bx) = (a_I, b_J)$ as needed. ■

Exercise 3.1: secretly part of 11.6 and 10.17(?)

Prove that every left primitive ring is prime, and that every semiprimitive ring is semiprime.

Proof.

Suppose R is left primitive. Let ${}_RM$ be a faithful simple module. For any non-zero ideal $I \subseteq R$, note that $IM = M$, since M is faithful and simple. Given two non-zero ideals $I, J \subseteq R$, we therefore have:

$$(IJ)M \;=\; I(JM) \;=\; IM \;=\; M$$

Therefore $IJ \neq 0$, so R is prime.

Now, suppose R is semiprimitive. Consider the nilradical $\sqrt{(0)}$; we want to show that $\sqrt{(0)} = (0)$. Indeed, recall that $\sqrt{(0)}$ is nil, and that any nil ideal is contained in $J(R)$. But since $J(R) = (0)$, we therefore have that $\sqrt{(0)} = (0)$, so (0) is semiprime as needed. ■

Exercise 4.1

Prove that if R is a commutative ring, then R is a domain iff R is prime. Give an example of a noncommutative prime ring which is not a domain.

Proof.

See Exercises 5 and 7 of the Section 10 worksheet (Lam Exercises 10.0 and 10.3).

Non-example: wip

■

Exercise 5.1: secretly part of Theorem 10.11

Let R be a ring and let I be an ideal of R. Prove that the following are equivalent:

i) I is a semiprime ideal of R, i.e. if $J^2 \subseteq I$ then $J \subseteq I$ for all ideals J.

ii) I is an intersection of prime ideals.

Proof.

($\Longleftarrow$) Suppose $I = \bigcap U$ is the intersection of prime ideals. If $J^2 \subseteq I$, then for each U:

$$J^2 \subseteq U \implies J \subseteq U$$

So $J \subseteq I$.

($\Longrightarrow$) Suppose I is a semiprime ideal. We show that $I = \sqrt{I}$. Clearly, $I \subseteq \sqrt{I}$ by definition. Now, if $a \notin C$, then $N := R \setminus C$ is an n-system containing a. Then there exists an m-system $M \subseteq N$ containing a, which is disjoint from C. Thus, $a \notin \sqrt{C}$.

NOTE: uses a lemma i havent written, see lemma 10.10

$\blacksquare$

Exercise 6.1: Subdirect Products

Let $\{S_i\}_{i\in I}$ be a family of rings indexed by some (possibly infinite) set I, and let S denote their direct product:

$$S \;=\; \prod_{i\in I} S_i \;:=\; \{(s_i)_{i\in I} : s_i \in S_i \quad \text{for all } i \in I\}$$

A subring R of S is said to be **full** if the projection $\pi_i|_R : R \to S_i$ is onto for all $i \in I$. A **subdirect product** of the S_i's is any ring which is isomorphic to a full subring of S.

Prove that R is a subdirect product of the S_i's if and only if there is a family of surjective ring homomorphisms $\varphi_i : R \to S_i$ such that $\bigcap_{i\in I} \ker(\varphi_i) = \{0\}$.

Proof.

($\Longrightarrow$) Suppose R is a subdirect product of the S_i's. Then, $R \simeq S' \subseteq S$ for some full subring S'; denote by φ this isomorphism. Then, consider the family of ring homomorphisms $\{\varphi_i\}$ given as:

$$\varphi_i : R \longrightarrow S \qquad \varphi_i \;=\; \pi_i \circ \varphi$$

where π_i is the i-th coordinate projection. Clearly these are surjective ring homomorphisms, so suppose $r \in \ker(\varphi_i)$ for all i. Letting $s := \varphi(r)$, we have:

$$\varphi_i(r) = (\pi_i \circ \varphi)(r) = \pi_i(\varphi(r)) = \pi_i(s) = 0 \quad \Longrightarrow \quad s_i = 0$$

Since this holds for all i, we get that $s = \varphi(r) = 0$, so $r = 0$ as needed.

($\Longleftarrow$) Suppose $\{\varphi_i\}$ is as given. Define the map φ as follows:

$$\varphi : R \longrightarrow S' =: \operatorname{im}(\varphi) \subseteq S \qquad \varphi(r) = (\varphi_i(r))_{i\in I}$$

It is easy to see that φ is indeed a ring homomorphism; by definition it is surjective. It is injective since by assumption:

$$r \in \ker(\varphi) \quad \Longrightarrow \quad \varphi_i(r) = 0 \quad \Longrightarrow \quad r \in \ker(\varphi_i) \qquad \text{for all } i$$

That is, $\ker(\varphi) = \bigcap_i \ker(\varphi_i) = (0)$. It remains to show that S' is a full subring.

Indeed, let $s_i \in S_i$. Then, there is $r \in R$ such that $\varphi_i(r) = s_i$. Let $s = \varphi(r)$. Then:

$$\pi_i(s) \;=\; \pi_i((\varphi_i(r))_i) \;=\; s_i$$

Thus, each π_i is surjective on S', so it is full as needed. ■

Exercise 6.2

Show that $\mathbb{Z}$ is a subdirect product of the family $\{\mathbb{Z}/n\mathbb{Z} : n \in \mathbb{N}\}$.

Proof.

For any $n \in \mathbb{N}$, consider $\varphi_n : \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z}$ as the canonical projection map. This is evidently a surjective ring homomorphism, so we show that $\bigcap_{n\in\mathbb{N}} \ker(\varphi_n) = \{0\}$.

Let $m \in \bigcap_{n\in\mathbb{N}} \ker(\varphi_n)$. Then for any fixed n, we have that:

$$m \in \ker(\varphi_n) \implies m \in n\mathbb{Z} \implies n \mid m$$

Since this is true for all $n \in \mathbb{N}$, we must necessarily have that $m = 0$. Since each φ_n is a ring homomorphism, we have thus shown both set inclusions. ■

Exercise 6.3

Prove that a ring R is semiprime if and only if R is a subdirect product of prime rings.

Proof.

($\implies$) Suppose R is semiprime. Consider the family of prime rings $\{R/P_i\}_{i\in I}$ where $\{P_i\}_{i\in I}$ is the family of all prime ideals of R. By part i), it suffices to consider the family of quotient maps $\{\pi_i : R \longrightarrow R/P_i\}$. Clearly:

$$\bigcap_{i\in I} \ker(\pi_i) \;=\; \bigcap_{i\in I} P_i$$

Since R is semiprime, $(0) \subseteq R$ is the intersection of prime ideals for some collection of prime ideals. Since this collection is contained in the family of *all* prime ideals of R, the above intersection is thus trivial as needed.

($\Longleftarrow$) Suppose R is a subdirect product of prime rings S_i. Let φ denote the given isomorphism and suppose $aRa = (0)$ for some $a \in R$. Let $b = \varphi(a)$ and $S' = \mathrm{im}(\varphi)$. Then, for any fixed $s \in S'$, we have:

$$aRa = (0) \implies bS'b = (0) \implies b_i s_i b_i = 0 \implies b_i = 0 \quad \text{for all } i$$

Thus, $b = \varphi(a) = 0$, so $a = 0$ as needed. ■

The structure of primitive artinian rings

Let M be a semisimple right R-module, and let $\Delta := \mathrm{End}(M_R)$ be the ring of R-module endomorphisms of M. For these exercises, assume the Jacobson density theorem in the following form: *if $\varphi : {}_\Delta M \to {}_\Delta M$ is an endomorphism of M as a left Δ-module, and $x_1, \ldots, x_n \in M$ is a finite list of elements of M, then there exists $r \in R$ such that $\varphi(x_i) = x_i r$ for all $1 \leq i \leq n$.*

Exercise 7.1

Our first goal is to find conditions under which the natural map $\Phi : R \to \mathrm{End}({}_\Delta M)$ is an isomorphism.

Prove that if R is artinian, then ${}_\Delta M$ is finitely generated.

Proof.

Recall that since R is artinian, M_R is finitely generated by, say, $\{m_1, \ldots, m_n\}$. Thus, for any $m \in M$, there exists $r_1, \ldots, r_n \in R$ such that:

$$m = \sum_{i=1}^{n} m_i r_i$$

For each r_i, we can define $\varphi_i \in \Delta$ as:

$$\varphi_i(m) = m r_i \implies m = \sum_{i=1}^{n} m_i r_i = \sum_{i=1}^{n} \varphi_i m_i$$

So ${}_\Delta M$ is in fact generated by the same $m_1, \ldots, m_n$ as M_R, as needed. ■

Exercise 7.2: secretly corollary 11.17

Use the Density Theorem to show that if ${}_\Delta M$ is finitely generated, then Φ is onto.

Proof.

Suppose ${}_\Delta M$ is finitely generated by $m_1, \ldots, m_n$. By the Density Theorem, for any $\varphi \in \mathrm{End}({}_\Delta M)$, there exists $r \in R$ such that $\varphi(m_i) = m_i r$ for all i. Therefore:

$$mr = \sum_{i=1}^{n} (a_i m_i) r = \sum_{i=1}^{n} a_i (m_i r) = \sum_{i=1}^{n} a_i \varphi(m_i) = \varphi\left(\sum_{i=1}^{n} a_i m_i\right) = \varphi(m)$$

Thus, Φ is indeed surjective. ■

Exercise 7.3

Prove that if M_R is faithful, then Φ is one-to-one.

Proof.

Indeed, if M_R is faithful, then $\mathrm{Ann}_R(M) = (0)$. If $r \in \ker(\Phi)$, or in other words, if $\Phi(r) \in \mathrm{End}({}_\Delta M)$ is the zero map, then by definition of Φ, $Mr = 0$. But then $r \in \mathrm{Ann}_R(M) = (0)$, so $r = 0$ as needed. ■

Exercise 7.4: Schur's Lemma

Our second goal is to interpret $\mathrm{End}({}_\Delta M)$ as a matrix ring over a division ring.

Prove that if M_R is simple, then Δ is a division ring.

Proof.

Let $\varphi \in \Delta$ be non-zero. Since $\ker(\varphi)$ and $\mathrm{im}(\varphi)$ are both submodules of M_R and $\varphi \neq 0$, we must have that:

$$\ker(\varphi) = \{0\} \qquad \mathrm{im}(\varphi) = M_R$$

So φ is an isomorphism, and thus invertible in Δ. ■

Exercise 7.5

Prove the general fact that if D is a division ring and N is a finitely generated left D-module, then $\mathrm{End}({}_D N)$ is isomorphic to the matrix ring $M_n(D)$.

You may freely use the fact that modules over division rings behave exactly like vector spaces. That is: every module over a division ring has a basis, any two bases have the same cardinality, and linear mappings are determined by their action on a basis. Thus every finitely generated D-module is isomorphic to ${}_D D^n$ for some $n \geq 0$.

Proof.

Given $A \in M_n(D)$, define $\varphi_A \in \mathrm{End}({}_D N)$ as $\varphi_A(x) = Ax$, where we consider $x \in N \simeq D^n$ with $n = \dim({}_D N)$. We show the map Φ given by $A \mapsto \varphi_A$ is an isomorphism.

Indeed, fix some basis $v_1, \ldots, v_n$ of N. If $\varphi_A = 0$ for some A, then for each basis vector v_i:

$$\varphi_A(v_i) = Av_i = 0 \implies A = 0$$

since $Av_i = 0$ for all i, so Φ is injective. It is also surjective, as any $\varphi \in \mathrm{End}({}_D N)$ is determined by its action on the basis elements v_i. Thus, $\varphi = \varphi_A$, where the i-th column of A is given by the coordinates of $\varphi(v_i)$. ■

Exercise 7.6

Conclude that if R is a left primitive artinian ring, then R isomorphic to a matrix ring $M_n(\Delta)$, where $n \geq 1$ and Δ is a division ring.

Proof.

Since R is left primitive, there exists a simple faithful module ${}_R M$. Notice that we can consider M as a right R^{op}-module, where for $r \in R$ and $m \in M$, we let:

$$m \cdot r' \;=\; r \cdot m$$

where $r' \in R^{\mathrm{op}}$ is the element opposite of r, notated for clarity. Denote $S := R^{\mathrm{op}}$; it then follows that M_S is a simple faithful module.

Let $\Delta = \mathrm{End}(M_S)$, and let $N = \mathrm{End}({}_\Delta M)$. Since R and thus S is artinian, then by the above work, we have that:

$$S \simeq N \simeq M_n(\Delta) \quad \Longrightarrow \quad R \simeq (M_n(\Delta))^{\mathrm{op}} \simeq M_n(\Delta^{\mathrm{op}})$$

Since the opposite ring of a division ring is evidently still a division ring, the result then follows. ■

Exercise 8.1

Prove that if R is artinian, then every primitive ideal of R is a maximal two-sided ideal.

Proof.

Let $I \subseteq R$ be a primitive ideal; then R/I is a primitive ring. Since R is artinian, then R/I is as well, and thus simple. Therefore, I is maximal. Since annihilators are two-sided and I is an annihilator of some simple R-module, it follows that I is two-sided as needed. ■

Exercise 8.2

Let R be semisimple. Prove that R has at most finitely many primitive ideals.

Proof.

Since R is semisimple, we can write:

$$R \simeq I_1 \oplus \cdots \oplus I_n$$

where each $I_j \subseteq R$ is a minimal two-sided ideal. It follows that the maximal ideals of R are precisely of the form:

$$I_1 \oplus \cdots \oplus \widehat{I_k} \oplus \cdots \oplus I_n$$

for some k, where the hat indicates omission. Evidently, there are only finitely many such ideals, so by the previous part, the result follows. ■

Exercise 9.1: Artin-Wedderburn Theorem

Combine (almost) everything above to prove that if R is an artinian semiprimitive ring, then R is isomorphic to a finite direct product of matrix rings.

10 The Jacobson Commutativity Theorem

currently just going off of the discord msgs, actual exercises may change

Theorem 10.1: Jacobson Commutativity Theorem

Let R be a ring such that for each $a \in R$, there exists some number $n \in \mathbb{N}$ such that $a^n = a$. Then R is commutative.

Exercise 10.2

Prove it in the case that R is a division ring.

Exercise 10.3

Prove it in the case that $R = M_n(\Delta)$ is a matrix ring over some division ring Δ.

Exercise 10.4

Prove it in the case that R is semisimple. Hint: Use AWT.

Exercise 10.5

Prove it in the case that R is semiprimitive. Hint: Use JDT to reduce to when R is also artinian.

Exercise 10.6

Prove it for any ring R. Hint: By the previous exercise, $R/J(R)$ is commutative.